# BIA 568-B Assignment 2 AI Knowledge Assistant Memo

**From:** [your name]

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## 1 Find handbook information

*What changed for each of Q01–Q06 when RETRIEVAL_K increased from 1 to 3? Use the retrieved and required passage IDs to explain.*

Raising RETRIEVAL_K from 1 to 3 raised required-source recall from 4/7 (57%) to 7/7 (100%) and complete questions from 2/5 to 5/5. Q01 (required S01) and Q05 (required S09) were already correct at K = 1. K = 3 only added unneeded passages (S03, S02 for Q01; S06, S05 for Q05). Q02 (required S04) failed at K = 1: BM25 ranked S11 (supplemental and spouse life) first. Its only match was the generic word "need", while S04 matched only "proof". Keyword search misses that "married" means "marriage" and "HR" means "Human Resources", and the longer S04 passage was penalized for its length. S04 came second. K = 3 recovered it (S11, S04, S01). Q03 (required S07, S06) and Q04 (required S03, S10) each need two topics, so one passage could never be enough. K = 1 returned only S07 (the copay) and only S03 (the start date). K = 3 added S06 (urgent care is not subject to the deductible) and S10 (1x earnings up to $150,000). Q06 has no required passage. At both settings BM25 still returned keyword matches (S14 commuter at K = 1; S14, S10, S08 at K = 3), none of which mention home internet. A larger K therefore gives the model more off-topic text it could misuse. The tradeoff: K = 3 finds every required passage but adds roughly one or two irrelevant passages per question.

## 2 Compare assistant configurations

*Which configurations scored best on Q01–Q05, and which responded fastest? Use the quality and time tables. For Q03 or Q04, explain how the answers changed. For Q06, explain how the assistant handled missing information.*

B (All 14 passages) and D (Top 3 passages) tied for best quality. Both scored a perfect 2.0 mean on Q01–Q05 and 2 on Q06. C (Top 1) averaged 1.4 and A (No handbook) 1.0. A was fastest, with a 0.233 s median response, followed by C (2.049 s), D (4.627 s) and B (10.116 s). Most of that gap was free-tier quota waiting (B 47.65 s total, D 26.71 s, C 10.07 s, A 0 s), not model time. Median API time was 0.23–0.27 s for A, C and D but 0.467 s for B, which sent about 1,847 input tokens per answer against D's 588. Q04: A only deferred to HR (score 1). C retrieved just S03, so it gave the date-of-hire start but said the handbook does not specify the amount (score 1). B and D had S03 and S10 and gave both facts, date of hire and 1x annual earnings up to $150,000 (score 2). Q06: A, B and D said they had no information on home internet and pointed to HR (score 2). C, given only the commuter passage S14, claimed "Stevens does not cover home internet expenses." That is an invented policy, and the judge scored it 0.

## 3 Try the assistant

*Give the two questions you tried. Could an employee use each answer? Describe one useful feature or one problem you noticed.*

I asked "Can my 24-year-old stay on my insurance?" (U05) and "Does Stevens help pay for grad school?" (U06). The default referral question was also logged as U01–U04. An employee could not use the U05 answer. BM25 retrieved S11, S10 and S03, all life-insurance passages matched on the word "insurance", and missed S02 (dependent children). The model then answered about adding a "24-year-old spouse" to supplemental life insurance. The correct answer is that a child stays on medical coverage until the end of the year they turn 26. The U06 answer was usable. It cited S12 and correctly described Tuition Aid of up to $6,000 after one year of service, though it dropped the full-time requirement. Problem noticed: the citation check passed U05 because S11 was a real supplied ID. It confirms that a source exists, not that the source answers the question.

## 4 Inspect the workflow

*Which workflow step took the longest? Explain how retrieval and the answer model contributed to one of your questions.*

The longest step was "Wait for allowance", the Groq free-tier quota wait, at 4.19 s for the referral question (U01). Every local step on the Colab CPU (loading passages, BM25 search, building the prompt, checking the reply) took under 0.001 s. For my two questions there was no quota wait, so "Generate answer" on Groq was the longest step (0.27 s for U05, 0.42 s for U06). For U06, BM25 ranked S14 (commuter), S10 (basic life) and S12 (educational assistance) in 0.0002 s. The useful passage was only third. S14 led because it matched "help" and "pay", while S12 matched only "Stevens"; "grad" and "school" matched nothing because the handbook says "tuition" and "higher learning". The answer model then did the real work: it ignored S14 and S10, drew the $6,000 Tuition Aid and one-year-of-service details from S12, and cited only S12. Retrieval narrows the context cheaply, but the answer depends on the right passage making the top 3.

## 5 Recommend a configuration

*Which configuration would you recommend? Support your choice with quality and timing result. Name one improvement before launching the assistant.*

I recommend D (Top 3 passages). It matched B's quality, with a 2.0 mean on Q01–Q05 and 2 on Q06, while sending about one third of the input (588 vs 1,847 tokens). It also had a faster median response (4.627 s vs 10.116 s) and less quota waiting (26.71 s vs 47.65 s). C was cheaper but averaged only 1.4 and invented a "no internet benefit" policy on Q06. A was fastest but could not answer any handbook question. D also handled missing information correctly by saying so and referring the employee to HR. Improvement before launch: U05 showed that keyword search fails when an employee's words differ from the handbook's ("24-year-old" vs "dependent children"). I would add semantic (embedding) retrieval alongside BM25, plus a check that defers to HR when no retrieved passage actually covers the question. A paid Groq tier would also remove most of the quota wait.